This week I submitted my thesis. I have never written anything longer than eight thousand words and most of the academic writing I've done was part of a larger group working on a paper together. I followed the framework that my research supervisor at King's introduced me to: fractal writing [1].

Starting with an overall concept for the introduction "Attention allocates limited resources between competing visual representations according to the statistics of the environment".

There is already quite a lot contained in this sentence.

Attention...
What is attention? What does it mean to allocate resources? How does attention alter the representation of a stimulus? How can attention be directed towards one stimulus rather than another?

"limited neural resources"...
What are the resources being allocated? Why are neural resources limited? What does a limited-resource model of perception predict? How can we measure the allocation of these resources?

"between competing visual representations"...
What happens when multiple stimuli have to be represented simultaneously? How does competition between stimuli affect their representations? Does representing one stimulus reduce the resources available for another? How does this relate to visual working memory?

"according to the statistics of the environment"...
Why should the environment influence how neural resources are allocated? What does efficient coding mean? Why are some visual features represented more precisely than others? What does the efficient coding hypothesis predict about the representation of orientation? Could attention amplify these existing differences in representation, or compensate for them?

Then expanding this line into a paragraph might result in the following.

Attention dynamically allocates limited neural resources between competing visual representations according to the statistics of the environment. Efficient coding models traditionally designed for single stimulus encoding propose that neural resources are distributed in accordance to natural scene statistics. When multiple visual stimuli compete for representation, attention may therefore alter how these limited resources are distributed between them, potentially reinforcing existing differences in sensory precision or compensating for poorly represented stimuli.

I found this preferable to writing in sections like introduction, methods, results and discussion. Those sections make sense when the work is finished, but they can impose an artificial order on the process of producing it. And these sections will naturally emerge anyway. Fractal writing instead allows the big picture to remain at the centre.

I also found that it helped with blocks. It is almost like increasing the surface area of your substrate. Instead of trying to generate one sentence after another, there are lots of little starting points available. If I ran out of ideas on one topic, I could simply pick up somewhere else. I could expand a definition, explain an assumption, follow up a reference, add an example, or develop an implication of something I had already written. I also tried to keep my fingers on the keyboard and continue typing. To keep it flowing, leaving sentences half finished and details missing is perfectly acceptable.

This style of writing lends itself to multiple passes. Create material, decide what it actually means, make it clearer, add references, remove, reposition. Much of my thesis was produced by writing sentences that I knew were not very good, leaving myself specific instructions about what was missing, and gradually turning those instructions into something coherent. And perhaps that is a better description of how the thesis was actually written. It wasn't written from beginning to end. It was assembled.

[1] Thank you to Dr Andrew Meso who has always encouraged forming good writing habits.